# MUHAMMAD AQIB SHER

## AI SYSTEMS ENGINEER | FULL-STACK, CLOUD & EDGE AI

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## PROFILE

AI Systems and Full-Stack Engineer with 5+ years of software development experience spanning AI-powered products, cloud and edge inference, backend platforms, computer vision, real-time systems and XR. Currently leads AI, XR and full-stack engineering at Suave Solutions, working across architecture, model integration, cloud infrastructure, deployment, performance optimization and production delivery. Experience includes deploying Cosmos Reason 3 inference workflows on NVIDIA DGX Spark, provisioning GPU-backed CloudXR infrastructure, building camera-driven AI analysis pipelines, and delivering AI-enabled web and learning platforms.

## CORE TECHNICAL SKILLS

**AI & intelligent systems:** LLMs, VLMs, AI copilots, RAG, multimodal workflows, voice agents, model APIs, inference integration
**Edge AI & inference:** NVIDIA DGX Spark, Cosmos Reason 3 deployment, edge inference workflows, real-time inference
**Cloud & GPU infrastructure:** AWS, Amazon Bedrock, cloud deployment, GPU instance provisioning, CloudXR, infrastructure troubleshooting
**Backend & web:** Node.js, REST APIs, React, React Native, Tailwind CSS, authentication, API integration, database-backed applications
**Computer vision & video:** IP cameras, RTSP, Re-ID, video processing, activity analysis, surgical-video labeling and step extraction
**Mobile & XR:** Kotlin, iOS/Android workflows, Unity, C#, AR/VR/MR, OpenXR, Quest, SBS video playback in VR
**Engineering & leadership:** System architecture, technical planning, debugging, optimization, production delivery, mentoring, stakeholder communication

## PROFESSIONAL EXPERIENCE

### Project Team Lead | AI, XR & Full-Stack Engineering

*Suave Solutions | Apr 2024 – Present | Lahore, Pakistan*

- Lead engineering across AI systems, computer vision, cloud infrastructure, real-time applications and spatial computing.
- Build AI-enabled web applications and backend services with Node.js, REST APIs and AWS deployment.
- Design and integrate LLM/VLM capabilities, RAG workflows, real-time APIs and intelligent application experiences.
- Develop camera-based AI pipelines using IP cameras, RTSP streams, computer vision and Re-ID for human activity and task analysis.
- Deploy Cosmos Reason 3 inference workflows on NVIDIA DGX Spark edge-computing hardware, including real-time inference experiments.
- Provision rented NVIDIA RTX PRO 6000-class GPU infrastructure for CloudXR streaming and troubleshoot GPU compatibility, networking and delivery to VR headsets.
- Deliver AI copilot, voice-agent, mobile/XR and video-learning workflows across connected applications and services.
- Own architecture, integration, deployment, debugging, optimization and delivery; coordinate developers and technical stakeholders.

### Senior Unity VR/AR Developer

*Symtera Technologies Pvt. Ltd. | Aug 2023 – Mar 2024 | Lahore, Pakistan*

- Developed VR/MR applications, including healthcare and bioscience simulation and training systems.
- Built interactive training workflows, API integrations, authentication and UI-driven content/library features.
- Optimized VR scenes and runtime performance; owned MR functionality through implementation and delivery.
- Implemented multiplayer features and adapted XR functionality across multiple headset platforms.

### Game Developer

*Abstron Private Limited | Sep 2021 – Aug 2023 | Lahore, Pakistan*

- Built mobile and VR applications across gameplay, UI, interaction systems and application architecture.
- Developed VR systems from the ground up, including gameplay mechanics, interactions and custom controller systems.
- Integrated APIs, external services and multiplayer features; maintained products and delivered updates.
- Optimized C# code and runtime performance while collaborating with 3D and animation teams.

### Web Developer Trainee

*XZSol Pvt. Ltd. | Aug 2021 – Sep 2021 | Pakistan*

- Supported server-side web application development, API/backend integration, full-stack workflows and database design tasks.

## SELECTED ENGINEERING WORK

**Cosmos Reason 3 on NVIDIA DGX Spark:** Deployed Cosmos Reason 3 on edge-computing hardware and implemented model inference workflows, including real-time inference.

**CloudXR GPU Streaming:** Worked with rented NVIDIA RTX PRO 6000-class cloud GPUs for CloudXR streaming and VR deployment, including infrastructure setup and compatibility/network troubleshooting.
**AI-Powered Surgical Video Learning:** Built workflows for managing learning content, processing large surgical and robotic-surgery videos, labeling content and deriving step-by-step event descriptions; supported SBS video playback in VR.
**AI / Computer Vision Pipeline:** Integrated camera streams, RTSP, computer vision, Re-ID and AI reasoning for real-time activity analysis.
**AI Cloud Platform & RAG:** Worked on end-to-end cloud AI platform architecture using Amazon Bedrock and RAG, connected to web application and backend workflows.

## ADDITIONAL ENGINEERING EXPERIENCE

**AI copilot systems:** Develop real-time AI/VLM copilot workflows using smart-glasses camera input for contextual guidance and assistance. Integrate multimodal model capabilities, application APIs and voice interaction into user-facing intelligent workflows.
**Backend and cloud engineering:** Build backend services and API integrations using Node.js and REST APIs, with AWS-based deployment and supporting cloud infrastructure. Design end-to-end cloud AI platform architecture, including Amazon Bedrock-based workflows and retrieval-augmented generation (RAG). Work across service integration, authentication, data flow, deployment, debugging and production support.
**Physical AI and computer vision & human activity analysis:** Develop camera-driven AI systems combining IP cameras, RTSP streams, computer-vision models, Re-ID and AI/VLM reasoning to interpret activities and task performance. Connect camera inputs, AI processing and application workflows to support human activity analysis and task validation.
**AI-powered learning and surgical-video intelligence:** Develop an AI-powered learning platform with user accounts, workflow management and a structured video-learning experience. Process large surgical and robotic-surgery videos, organize and label video content, and derive step-by-step descriptions of actions and events for learning and review. Support SBS/side-by-side video workflows and immersive playback of SBS video in VR, connecting web-based learning workflows with XR experiences.
**CloudXR and GPU infrastructure:** Provision and configure rented cloud GPU infrastructure, including NVIDIA RTX PRO 6000-class instances, for CloudXR and VR streaming workflows. Deploy and troubleshoot GPU-hosted VR streaming, investigating runtime/GPU compatibility, network paths, firewall configuration and low-latency delivery to Quest headsets.
**Web applications:** Build AI-enabled web applications and product workflows using Node.js, React and Tailwind CSS, integrating backend APIs, AI services and user-facing interfaces. Work across platform architecture, workflow orchestration, API integration and deployment.
**iOS, Android and native Android Kotlin apps for AR devices:** Experience across mobile application workflows, including iOS/Android delivery and native Android development with Kotlin for AR-device use cases. Integrate mobile applications with device capabilities, backend APIs and connected AI/XR experiences.
**End-to-end system design and engineering:** Translate broad requirements into system architecture spanning clients, APIs, AI models, cloud services, edge devices and operational workflows. Own technical implementation across integration, debugging, optimization, deployment and delivery while coordinating with engineers and stakeholders.
**POS systems:** Contribute to point-of-sale application work, connecting application workflows with supporting backend and user-interface functionality.
**Low-cost voice-to-voice agents:** Build voice-to-voice AI agent workflows with a focus on practical latency, service integration and cost-conscious implementation.
**Unity AR/VR/MR apps and games:** Develop immersive AR, VR and MR applications and games using Unity and C#, including training simulations, interaction systems, multiplayer features and cross-device support. Optimize real-time performance and deliver XR features from implementation through integration and release.
**LLM deployment on cloud GPUs:** Work with self-hosted/open-weight LLM deployments on GPU-backed cloud infrastructure, including model families such as Qwen, Gemma and other models. Configure inference environments and evaluate deployment approaches for AI applications, balancing model capability, hardware constraints and runtime needs.

## EDUCATION

BS Computer Science — COMSATS University Islamabad, 2017–2021
F.Sc. Pre-Engineering — Govt. Faridia College, Pakpattan, 2015–2017

## LANGUAGES

English · Urdu · Punjabi